Jian Feng

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Education

University of Waterloo Sep 2023 - Aug 2027
Bachelor's degree, Computer Science, Finance, and Statistics *Waterloo, ON, Canada*

Skills

Programming & SQL:	Python, SQL, R, pandas
Data Science:	scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LoRA, time series
Data Platforms:	Airflow, dbt, BigQuery, PostgreSQL, MySQL, SQL Server
Analytics & BI:	statistical analysis, feature engineering, Excel, Power BI, Tableau, Looker Studio
Finance:	equity research, DCF/IRR, portfolio analysis, risk management, factor investing, derivatives, fixed income, credit

Experience

Manulife Financial Corporation Toronto, ON, Canada
Software Engineer *May 2026 – Aug 2026*

- Partnered with Canadian segment and technical teams to refine reusable ServiceNow workflows for **5K+ developers**, automating **1K+ daily operations**; integrated live AKS and certificate checks into a **Foundry LLM agent** to flag GitOps deployment mismatches.
- Replaced plaintext GitHub credentials with **GitHub App token exchange** (PyGithub), adding team-scoped validation, least-privilege permissions, and audit trails.

University of Waterloo Waterloo, ON
Research Assistant *Jan 2026 – Apr 2026*

- Fine-tuned four specialist **DeBERTa safety detectors** with LoRA; neutral-context training reduced medical/education false positives, with **sub-50ms inference** on a single T4.
- Built a learned, per-axis aggregator that beat fixed-threshold fusion (**80% F1, 95%+ end-to-end accuracy** on held-out Hugging Face data); red-teamed against Llama-3-8B on **5K prompts**.

Treasury Board Secretariat Toronto, ON, Canada
Software Developer *Sep 2025 – Dec 2025*

- Automated duplicate and invalid job-title cleanup in a **5K-firm public dataset**, reducing manual processing time **85%** and enabling peer-group and governance analysis.
- Migrated **30 Selenium tests across five suites** to Katalon on Azure DevOps CI/CD; the executive demo led two other QA teams to adopt the framework and cut their no-code onboarding.

GuangDong Tiptop Asset Management Co., Ltd China
Analyst *Jan 2025 – Apr 2025*

- Screened **500+ listed equities** on earnings quality, moat durability, and capital intensity, distilling the universe into a **weekly portfolio-manager shortlist**.
- Supported company diligence and three-to-five-year forecasts; built DCF/IRR/MOIC scenarios and contributed to investment-committee materials and sell/hold recommendations.

Lansi Institute of Artificial Intelligence in Medicine Remote
Data Analyst *May 2024 – Dec 2024*

- Built a reproducible genomic ETL pipeline for **100GB of multi-omics data** in an international type 2 diabetes collaboration; parallel preprocessing cut runtime **89%, from days to hours**.
- Applied XGBoost and SMR/MR² across a **68-study meta-analysis** to screen evidence and prioritize candidate drug targets for published research.

Projects

Most Efficient Portfolio | Canadian Home Bias

- Built a CAD-denominated mean-variance frontier from five regional equity ETFs over **275 months**; quadratic programs and a $\gamma = 3$ utility criterion put the optimal Canadian weight at **48%**, close to the **~50%** observed home bias versus Canada's **~3%** global market cap.
- Ran sensitivity across $\gamma = 1$–10 and the minimum-variance criterion, documenting corner solutions and error maximization.

Earnings Yield Factor | CRSP/Compustat

- Built a point-in-time trailing E/P factor on CRSP/Compustat (**1.3M stock-months**, 2001–2024); the equal-weighted Q5–Q1 portfolio earned **18.7%/yr** (Sharpe 1.03, Newey–West $t = 4.13$).
- FF5 spanning leaves a **14.9%/yr alpha** ($t = 5.11$, $R^2 = 0.50$); value-weighting halves the premium to 9.5% ($t = 1.80$).

CabStream ETL | NYC Taxi Analytics

- Built an Airflow pipeline to ingest NYC taxi data into GCS and BigQuery, transform it into dbt marts, and visualize trip patterns in Looker Studio.
- Defined reproducible GCP infrastructure with Terraform and added DAG logic tests and dbt freshness/uniqueness checks.

Achievements

- **3rd of 45 teams**, SAF Investment Research Challenge; finalist, 7th Annual SAF Strategy Case Competition; WorldQuant Brain Gold Level (**20 quality alphas; Research Consultant candidate**), Velocity Co-op Problem Award (top submission in cohort).
- As UX Research Hub Director, owned community growth and programming end-to-end: ran member research, secured a Google UXR guest speaker (**50 attendees**, **50 new members**), grew membership **32× (10→318)**, and managed event funding with the executive team.